

SEONGSAN, Korea, Republic of

WMO: 471880

Lat: 33.3867N

Lon: 126.8803E

Elev: 19

StdP: 101.09

Time Zone: 9.00 (E09)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	0.0	-8.2	1.9	3.2	-7.0	2.1	3.2	8.2	8.1	7.2	6.2	3.2	320	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	31.5	26.9	30.6	26.6	29.6	26.2	27.7	30.1	27.2	29.5	26.6	28.9	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	29.1	26.5	22.1	28.7	25.9	21.3	28.4	88.9	30.1	86.1	29.6	83.6	29.1	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.2	6.3	-2.6	32.8	1.7	1.1	-3.9	33.5	-4.9	34.2	-5.9	34.8	-7.1	35.6		
			-4.2	28.3	1.4	0.9	-5.2	28.9	-6.0	29.4	-6.8	29.9	-7.8	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	5.1	6.1	9.5	14.1	18.1	21.2	25.7	26.9	23.4	18.6	13.2	7.3
	DBStd	7.85	3.07	3.41	3.04	2.48	1.89	1.49	2.27	1.97	1.75	2.74	3.83	3.43
	HDD10.0	430	154	116	47	2	0	0	0	0	0	12	99	
	HDD18.3	1711	410	342	274	128	26	1	0	0	31	158	342	
	CDD10.0	2556	3	7	32	125	251	334	488	525	401	265	108	16
	CDD18.3	794	0	0	0	1	19	85	230	267	151	37	4	0
	CDH23.3	5551	0	0	0	2	44	129	1941	2687	683	63	2	0
	CDH26.7	1373	0	0	0	0	1	2	473	840	56	1	0	0
Wind	WSAvg	3.1	3.2	3.4	3.4	3.3	2.9	2.6	3.1	3.2	3.2	3.3	2.9	3.0
Precipitation	PrecAvg	1827	74	92	118	140	167	240	278	286	203	90	84	55
	PrecMax	3194	168	206	269	212	330	841	699	783	742	420	193	179
	PrecMin	1070	18	35	32	50	12	41	82	66	38	0	13	5
	PrecStd	455	42	51	61	44	91	164	176	177	162	97	49	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	16.6	18.8	21.9	26.2	26.7	32.4	32.8	29.2	26.1	23.2	19.0
		MCWB	13.0	15.2	14.2	15.9	17.9	20.8	27.0	26.8	23.8	23.5	18.9	17.2
	2%	DB	13.6	15.6	17.4	20.4	24.5	25.4	31.2	31.9	28.1	25.0	21.6	16.4
		MCWB	10.7	13.8	13.9	15.6	16.9	21.4	26.8	26.9	23.5	22.8	18.7	13.5
	5%	DB	11.8	13.3	16.4	19.4	23.3	24.7	30.2	31.2	27.2	23.9	20.4	14.7
		MCWB	9.1	10.4	13.6	15.0	17.1	21.5	26.5	26.8	23.4	20.6	17.7	11.6
	10%	DB	10.1	11.8	15.3	18.4	22.2	24.0	29.2	30.4	26.4	22.8	19.1	12.8
		MCWB	7.1	8.5	12.6	14.8	17.2	21.4	26.1	26.5	22.9	19.3	16.1	9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	16.0	16.3	18.3	21.1	24.5	28.4	28.7	25.9	24.4	20.7	17.6
		MCDB	15.1	16.4	17.1	20.2	22.3	25.2	30.9	30.7	27.9	25.5	22.2	18.8
	2%	WB	11.1	13.8	15.3	17.5	20.2	23.5	27.5	28.1	25.1	23.3	19.5	14.5
		MCDB	13.1	15.0	16.5	18.9	22.0	24.3	30.0	30.3	26.7	24.6	21.1	15.9
	5%	WB	9.4	11.3	14.5	16.8	19.5	22.8	27.0	27.5	24.4	21.8	18.0	12.0
		MCDB	11.7	12.8	16.1	18.1	21.5	23.8	29.4	29.8	26.1	23.5	20.1	14.6
	10%	WB	7.7	9.2	12.9	16.0	18.8	22.2	26.5	27.0	23.8	20.1	16.7	10.2
		MCDB	10.0	11.3	15.1	17.6	20.8	23.4	28.6	29.3	25.6	22.0	18.7	12.7
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	7.1	7.2	6.8	4.6	4.3	4.7	4.9	5.7	6.1	5.5
		MCDBR	8.1	7.8	8.7	8.5	9.2	6.0	5.7	5.9	5.7	5.8	6.7	7.2
		MCWBR	6.7	6.8	7.1	6.0	5.8	3.6	2.9	2.8	3.2	3.9	5.2	6.0
	5% WB	MCDBR	7.6	7.2	7.8	5.5	6.3	4.1	5.1	5.1	4.7	4.6	5.6	6.4
		MCWBR	6.9	7.8	7.5	5.1	4.9	3.1	3.0	2.8	3.3	4.1	5.2	5.9
Clear-Sky Solar Irradiance	taub	0.459	0.499	0.552	0.563	0.571	0.600	0.580	0.537	0.498	0.469	0.457	0.451	
	taud	1.968	1.904	1.792	1.820	1.855	1.876	1.976	2.106	2.153	2.100	2.054	1.996	
	Ebn at Noon	714	732	729	746	745	722	734	760	771	756	715	694	
	Edh at Noon	146	173	208	211	206	201	181	157	144	140	133	135	
All-Sky Solar Radiation	RadAvg	2.10	2.92	4.12	4.89	5.24	4.57	5.03	5.13	4.21	3.71	2.60	1.95	
	RadStd	0.24	0.36	0.33	0.41	0.60	0.56	0.72	0.52	0.41	0.37	0.29	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page